

RIVERSDALE, South Africa

WMO: 689260

Lat: 34.0958S

Lon: 21.2517E

Elev: 152

StdP: 99.51

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	4.8	-0.2	3.8	15.0	1.2	4.2	12.5	6.2	15.9	5.4	15.9	0.4	250	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	32.5	20.3	30.5	19.9	28.9	19.4	21.9	29.1	21.2	27.6	20.5	26.4	3.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.8	24.1	19.2	14.2	23.5	18.5	13.6	23.0	64.6	29.1	62.1	27.9	59.8	26.7	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	6.1	5.3	DB	1.8	38.1	1.2	1.7	1.0	39.3	0.3	40.3	-0.4	41.3	-1.3	42.6	(4)
			WB	1.4	23.7	1.2	0.5	0.6	24.0	-0.1	24.3	-0.8	24.6	-1.6	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	22.5	22.6	21.4	18.8	16.5	13.3	12.9	13.8	15.1	17.5	18.8	21.2
	DBStd	4.36	2.46	2.39	2.71	3.11	2.63	2.60	3.10	2.45	2.60	2.96	2.55	2.44
	HDD10.0	10	0	0	0	0	0	2	6	1	1	0	0	0
	HDD18.3	757	1	1	4	31	71	154	172	143	101	51	24	4
	CDD10.0	2874	389	354	354	263	200	100	97	118	154	234	264	347
	CDD18.3	579	132	122	101	44	12	2	4	2	4	26	38	92
	CDH23.3	3334	737	662	506	277	88	23	48	33	54	178	204	524
CDH26.7	984	227	200	147	98	17	7	9	7	17	58	51	147	
Wind	WSAvg	2.0	2.7	2.4	2.1	1.8	1.5	1.4	1.5	1.6	1.8	2.3	2.6	2.7
Precipitation	PrecAvg	445	22	35	35	57	28	39	38	45	32	50	38	33
	PrecMax	613	64	74	100	161	65	62	100	178	69	122	151	99
	PrecMin	287	1	6	4	6	2	9	8	6	6	7	4	0
	PrecStd	91	14	18	23	42	18	16	22	39	19	37	36	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	33.7	34.0	33.8	28.7	27.4	27.9	27.2	29.9	32.4	31.8	33.2
		MCWB	21.6	21.2	20.8	19.0	15.9	15.2	14.3	14.8	15.7	19.3	18.9	20.7
	2%	DB	31.4	31.0	30.1	29.0	25.6	22.6	24.0	23.2	24.7	27.9	27.8	30.1
		MCWB	20.9	20.7	20.3	18.7	15.2	13.2	13.0	14.2	15.1	17.7	18.1	20.1
	5%	DB	29.2	29.0	28.0	25.9	23.4	20.2	21.3	20.7	22.2	24.8	25.5	28.2
		MCWB	20.3	20.2	19.4	17.6	15.1	13.0	12.7	13.2	14.4	16.4	17.4	19.2
	10%	DB	27.4	27.2	26.1	23.7	21.5	18.4	18.7	18.9	20.3	22.6	23.7	26.4
		MCWB	19.6	19.5	18.9	16.9	14.8	12.5	12.4	12.9	13.8	15.9	16.6	18.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	22.8	22.0	20.1	17.9	16.1	15.3	16.7	17.7	20.5	20.3	22.2
		MCDB	30.9	31.6	29.8	29.6	23.7	20.1	23.0	22.4	27.1	28.9	28.1	30.5
	2%	WB	21.8	21.7	21.1	19.2	16.8	15.0	14.4	15.4	16.2	18.6	19.0	20.9
		MCDB	28.6	28.5	28.5	26.3	22.1	20.1	20.5	20.2	22.6	25.2	25.4	27.8
	5%	WB	21.1	20.9	20.3	18.4	16.0	14.1	13.6	14.4	15.2	17.4	18.1	20.0
		MCDB	27.0	26.4	26.1	24.2	21.1	18.6	18.7	19.0	20.7	23.0	23.8	26.2
	10%	WB	20.4	20.2	19.6	17.7	15.3	13.2	12.9	13.4	14.2	16.6	17.3	19.2
		MCDB	25.7	25.3	24.5	22.7	20.0	17.0	17.6	17.5	19.1	21.3	22.5	24.6
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.1	10.0	10.4	10.2	10.3	10.8	11.2	10.5	10.1	9.9	10.0
		MCDBR	13.0	13.2	12.9	14.1	12.9	13.4	15.3	14.9	14.4	14.5	13.5	13.7
		MCWBR	4.4	4.7	4.4	5.2	5.2	6.5	6.6	6.7	6.2	5.6	5.3	5.1
	5% WB	MCDBR	11.4	11.5	11.3	12.4	10.9	10.8	12.5	11.9	12.6	12.0	11.6	11.7
	MCWBR	4.2	4.4	4.2	5.1	5.1	5.8	6.5	6.1	6.1	5.1	4.9	4.6	
Clear-Sky Solar Irradiance	taub	0.360	0.362	0.349	0.326	0.312	0.300	0.303	0.327	0.339	0.361	0.349	0.357	
	taud	2.489	2.489	2.518	2.570	2.588	2.590	2.573	2.493	2.441	2.413	2.482	2.486	
	Ebn at Noon	977	954	929	898	858	843	859	881	924	944	981	984	
	Edh at Noon	115	111	101	85	74	69	73	90	107	119	115	116	
All-Sky Solar Radiation	RadAvg	7.61	6.81	5.55	4.13	3.07	2.62	2.89	3.71	5.04	6.22	7.30	7.83	
	RadStd	0.27	0.23	0.27	0.19	0.16	0.15	0.15	0.16	0.25	0.34	0.30	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page